

Class 11: Protein Structure Prediction with AlphaFold

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```
library(bio3d)

pdb <- read.pdb("hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v1.pdb")

pdb
```

```
Call: read.pdb(file = "hivpr_dimer_23119/hivpr_dimer_23119_unrelaxed_rank_001_alphafold2_multimer_v1.pdb")
```

```
Total Models#: 1
```

```
Total Atoms#: 1514, XYZs#: 4542 Chains#: 2 (values: A B)
```

```
Protein Atoms#: 1514 (residues/Calpha atoms#: 198)
```

```
Nucleic acid Atoms#: 0 (residues/phosphate atoms#: 0)
```

```
Non-protein/nucleic Atoms#: 0 (residues: 0)
```

```
Non-protein/nucleic resid values: [ none ]
```

```
Protein sequence:
```

```
PQITLWQRPLVTIKIGGQLKEALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYD
QILIEICGHKAIGTVLVGPTPVNIIGRNLLTQIGCTLNFPQITLWQRPLVTIKIGGQLKE
ALLDTGADDTVLEEMSLPGRWKPKMIGGIGGFIKVRQYDQILIEICGHKAIGTVLVGPTP
VNIIGRNLLTQIGCTLNF
```

```
+ attr: atom, xyz, calpha, call
```

Make a vector of input PDB file names that we can read into R

```
pdbfiles <- list.files("hivpr_dimer_23119/", pattern=".pdb", full.names=TRUE)
```